



9/1/2026

N + - Network Fundamentals & Building Networks.

Network devices.

TASK 1.

- ❖ Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.

➤ Home Network Diagram:

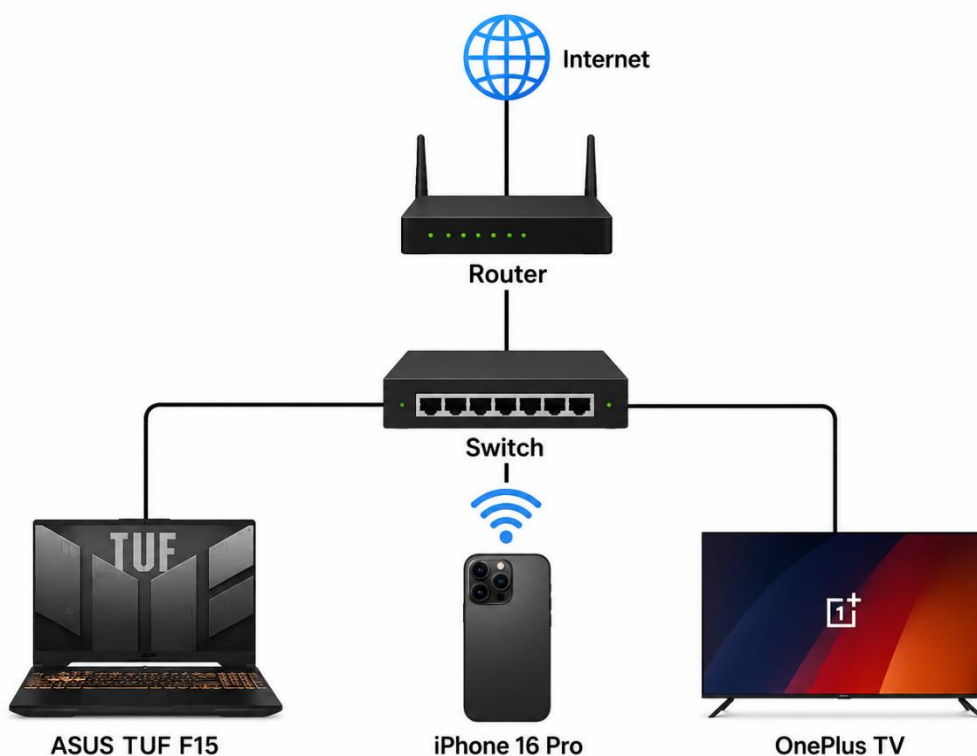


Figure : Simple Home Network Showing Router, Switch, and Connected Devices.

🌈 Connections:

- **Router → Switch:** Wired Ethernet connection
- **Switch → ASUS TUF F15:** Wired Ethernet connection
- **Switch → iPhone 16 Pro:** Wireless Wi-Fi connection
- **Switch → OnePlus TV:** Wired Ethernet connection

TASK 2.

❖ Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

➤ Comparison of Network Devices:

<u>Network Device</u>	<u>Real-World Use Case</u>
Router	Connects home devices to the Internet, allowing services like YouTube streaming and web browsing.
Switch	Connects multiple devices in a LAN, such as computers used for LAN gaming.
Hub	Connects multiple devices in a small basic network where data is shared with all connected devices.
Access Point	Provides Wi-Fi connectivity to devices such as smartphones and laptops for daily Internet use.

TASK 3.

- ❖ **Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).**

- **Data Travel from Phone to the Internet:**

When I order food on Zomato using my phone, the order data first travels through the Access Point, which connects my phone to the Wi-Fi network. The data then passes through a Switch, which forwards it within the local network. After that, the Router sends the data to the Internet and routes it to Zomato's servers. Zomato processes the order and sends the response back through the same network path to my phone.

TASK 4.

- ❖ **Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup, Think about devices like switches, powerline adapters, or WiFi extenders.**

- **Connecting a Gaming PC to the Internet:**

- ⌘ **Powerline Adapter:**

A powerline adapter uses the existing electrical wiring in the house to carry network data. One adapter is connected to the router, and another is placed near the gaming PC and connected to it using an Ethernet cable. This allows the PC to access the Internet without running a long Ethernet cable.

⌘ **Wi-Fi Extender:**

A Wi-Fi extender receives the Wi-Fi signal from the router and extends it to areas farther away. The gaming PC can be connected to the extender using an Ethernet cable, allowing the PC to access the Internet through the extended Wi-Fi connection.